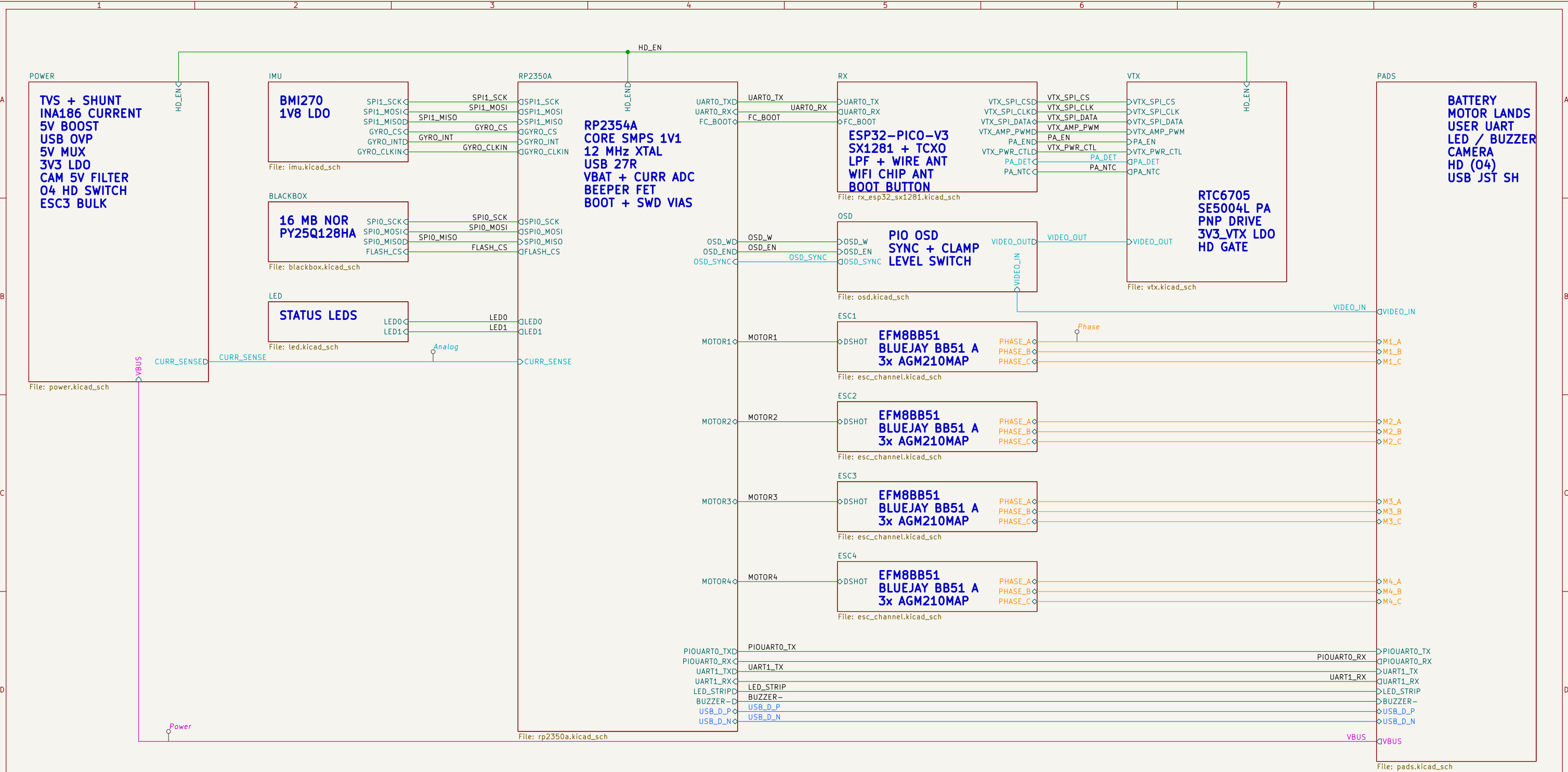




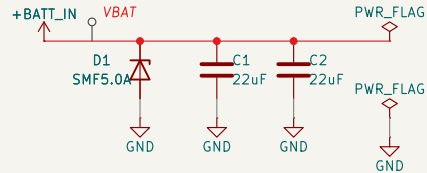
POWER TABLE

| Rail     | Source                      | Nominal    | Budget (spec 5)                             | Feeds                                  |
|----------|-----------------------------|------------|---------------------------------------------|----------------------------------------|
| +BATT_IN | battery pads J2 / J3 (1S)   | 3.0–4.35 V | 4x ESC + 2.0 A board                        | TVS, pad bulk, shunt R1                |
| +BATT    | shunt R1 0.5 m0hm           | cell       | ESC + boost in 1.4–3.0 A                    | ESC1–4, boost, 04 EN, VBAT ADC, PA_VCC |
| /VBUS    | USB J31 pin 4 (hier. net)   | 5 V        | 0.42 A, USB only                            | U24 OVP                                |
| +5V_USB  | U24 SGM40661 OVP            | 5 V        | as VBUS                                     | U3 VIN1, PR1 divider                   |
| +5V_BST  | U2 TPS61022 boost           | 5.15 V     | 0.70 A analog / 1.48 A HD (BEC 1.5 A cont.) | U3 VIN2, U5 04 switch                  |
| +5V      | U3 TPS2116 mux              | 5 V        | 0.70 A analog, about 0.25 A HD              | 3V3 LDOs, CAM filter, 5V pads          |
| +5V_CAM  | FB1 ferrite from +5V        | 5 V        | 0.12 A                                      | camera pad + plug                      |
| +5V_HD   | U5 TPS22810 (HD_EN, cell)   | 5 V        | 1.20 A 04 Lite                              | VHD pad                                |
| +3V3     | U4 LP5912–3.3               | 3.3 V      | 0.20 A (0.48 A Wi–Fi bench)                 | FC, IMU, NOR, OSD, RX, LEDs, INA186    |
| +3V3_VTX | U21 LP5912–3.3 (off in HD)  | 3.3 V      | 0.10 A typ, about 0.17 A at max PA          | RTC6705, PA drive, PA ref              |
| +1V8     | U12 TPS7A2018               | 1.8 V      | < 1 mA                                      | BMI270                                 |
| +1V1     | RP2354A core SMPS           | 1.1 V      | internal                                    | RP2354A DVDD                           |
| PA_VCC   | R67 selector, +BATT default | cell       | 0.65 A at 27 dBm                            | SE5004L PA (local, VTX)                |
| GND      | battery –                   | 0 V        | –                                           | all sheets                             |

Generated from hardware/sch\_contract.json by hardware/tools/sch\_root.py; do not edit by hand.  
Root wires carry the project-wide net names (root label = net /NAME). Motor phases are named by the ESC sheets (/ESCn/PHASE\_x). Rails are power symbols (table); VBUS is a hierarchical net.  
ESC1–ESC4 drive M1 rear right, M2 front right, M3 rear left, M4 front left (Betaflight quad–X); FC pin map: research/PINMAP.md.

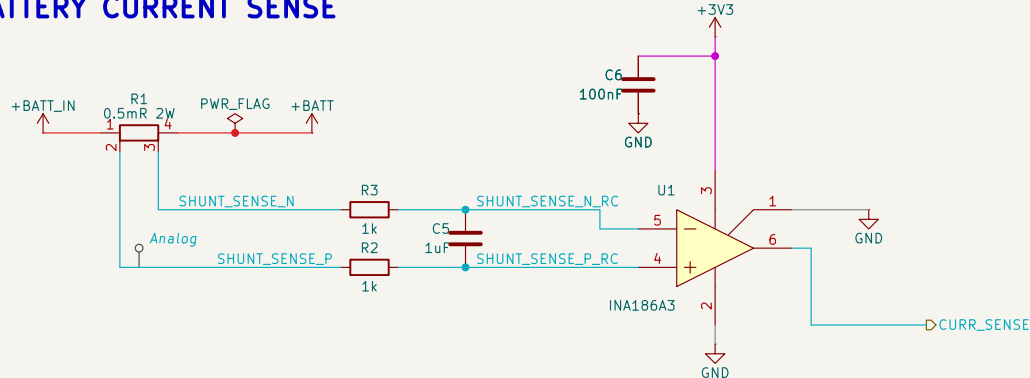
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## BATTERY INPUT AND TVS

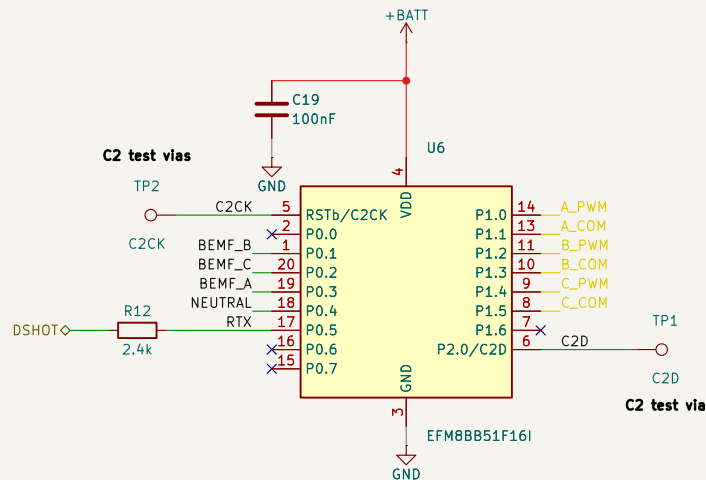


Source: OpenAIO/hardware/OpenAIO.kicad\_sch (battery entry), adapted.  
+BATT\_IN = battery pads J2 (+) / J3 (-) on the PADS sheet, 1S LiPo/LiHV 3.0–4.35 V; PWR\_FLAG here (the pads are passive). No reverse-polarity FET (spec 4.1: user-fitted pigtail, risk in README).  
D1 SMF5.0A: VBR 6.40–7.07 V, VC 9.2 V at 21.7 A, 200 W 10/1000 us; takes one motor's regen energy (~0.29 J per rated pulse) on a battery disconnect; fallback SMAJ5.0A (SMA, 400 W) if V3b exceeds it. It does not hold the 5.5–7 V parts (EFM8 VDD, TPS61022, TPS2116) in rating; Matrix-parity risk, spec 15.4, C1/C2 22 uF 16 V X5R 0603 at the pads; 9–13 uF effective each at 4.35 V. Hot plug of a 4.35 V pack can overshoot to 5.0–5.4 V (V3c); damping fallback ~100 uF polymer at the pads; PE–8 waits for V3c.

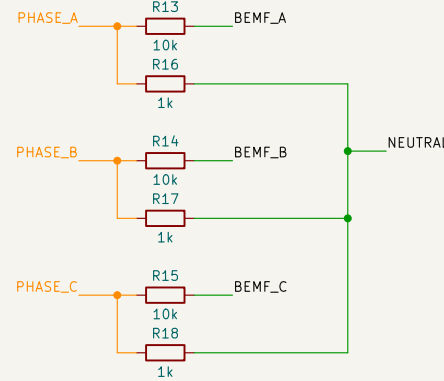
## BATTERY CURRENT SENSE



## ESC MCU: EFM8BB51, Bluejay BB51 layout A



## BEMF network: 10k series + 1k neutral star



Source: tinyPEPPER single\_esc.sch (CERN-OHL v1.2, wire trace in research 08): 10k series from each phase to its comparator input (BEMF\_A P0.3, BEMF\_B P0.1, BEMF\_C P0.2) and 1k from each phase to the virtual neutral (P0.4 V\_Mux, CMP0 negative). No leg to GND: at 1S the phase stays within the cell + one body-diode drop, and the 10k limit the input clamp current to  $< 0.1 \text{ mA}$ , 1k star, worst case one phase low and two high:  $2/3 \times 5.25 \text{ V} / 1\text{k} = 3.5 \text{ mA}$ , 12 mW per 0201 (50 mW part). P4-B: the netlist puts BEMF\_A/B/C and NEUTRAL on U6 pins 19/1/20 and 18 = A.inc A\_Mux 3 / B\_Mux 1 / C\_Mux 2 / V\_Mux 4.

ESC channel, one file placed 4x. The references drawn are ESC1's; the root sheet offsets them per instance (ESC2 U+1 Q+3 R+7 C+6 TP+2, ESC3 x2, ESC4 x3), matching bom\_plan.json:

ESC1 M1 rear right: U6 C19, hot loop C20 C23 C21, R12-R18, Q29-Q31, TP1/TP2

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ESC3 M3 rear left: U8 C31, hot loop C32 C35 C33, R26-R32, Q35-Q37, TP5/TP6

ESC4 M4 front left: U9 C37, hot loop C38 C41 C39, R33-R39, Q38-Q40, TP7/TP8

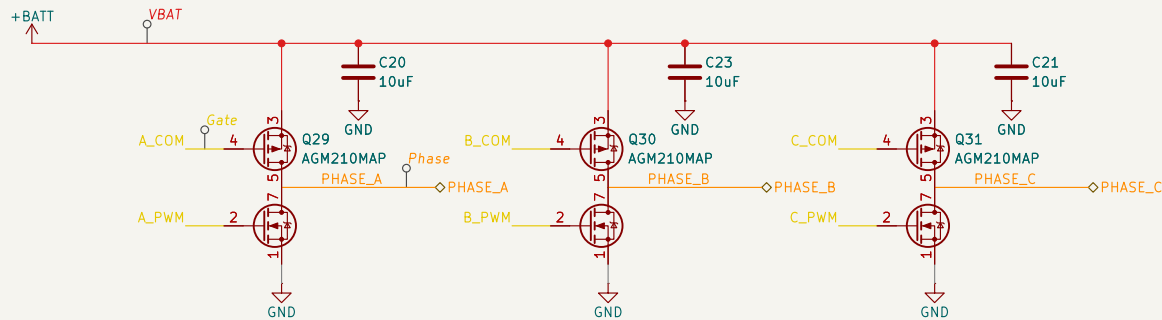
Pins: DSHOT (left, root net /MOTORn: FC GPIO25/24/23/22); PHASE\_A/B/C (right, local label + hierarchical label, net /ESCn/PHASE\_x, motor lands J4/J7/J10/J13). In-sheet nets: x\_COM, x\_PWM (Gate class), BEMF\_x, NEUTRAL, RTX, C2D, C2CK. Netclass flags: Phase (PHASE\_A), Gate (A\_COM), VBAT (+BATT rail): setup\_board.py patterns /ESC?/PHASE\_?, /ESC?/?\_COM, /ESC?/?\_PWM assign the rest.

Source: copied from OpenAIO hardware/esc\_channel.kicad\_sch (frames, block titles, C2 test points at the MCU pins, three-column BEMF block, power stage along the bottom) and adapted: AT32F421 + NSG2065Q + 6x DOY180N03T become EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, Matrix 1S 5IN1 II stage, D56), the 3S divider becomes the tinyPEPPER 1S star.

Power-up (P4-11, PINMAP F16): battery first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, tPOR  $\leq 390 \text{ us}$ ) and the Bluejay bootloader turns on the 70-220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1-2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV full), through  $\geq 72\text{k}$  (pull-up + R12), i.e.  $\leq 60 \text{ uA}$ ; 0.72 V over the 3.63 V unpowered-FT limit, current-limited. Accepted on Matrix parity (G473 FT pads about 4 V unpowered); V9 scopes the pad at power-up; fallback a 3.3 V-domain clamp or a DShot-path enable (not in rev1).

USB only: the PIO DShot idle-high back-feeds +BATT through the EFM8 I/O diode,  $\leq (3.3 \text{ V} - \text{VF}) / 2.4\text{k}$ , about 1 mA per channel; the EFM8s may cycle near POR, +5V stays on USB via the TPS2116 (10k fallback if V9 shows brown-out cycling).

## Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP (AGMSEMI, DS VER2.73 p1 top view: D23 rule, P3 symbol/footprint check): 1 S1 N source = GND, 2 G1 N gate = x\_PWM, 3 S2 P source = +BATT, 4 G2 P gate = x\_COM, 5/6 + EP D2 (P drain) and 7/8 + EP D1 (N drain) = PHASE\_x; the two drain pads are joined only by the phase island on the board (never split it). Gates straight from the EFM8: no gate resistors and no OR options (tinyPEPPER / Matrix II field record; bom\_plan v3 has none); the about 0.15 A gate-edge current for tens of ns is a transient above the EFM8's 50 mA per-pin abs max, accepted as on the Matrix II (spec 4.4). VGS  $\pm 12 \text{ V}$ , VDS 20 V abs max vs VDD  $\leq 5.25 \text{ V}$  regen peak. Hot loop (D77, P4 critique): one cap per AGM210MAP, C20 / C23 / C21 10 uF X6S 0402 from pin 3 S2 (+BATT) to pin 1 S1 (GND) at the cell's lead side, both pads via-in-pad, so the commutation loop closes at the pins (Matrix II: one cap per FET); the three are also the ESC +BATT bulk. Bluejay: first power-up A\_X\_10\_96 (about 200 ns; DS td(off) + tf = 58 ns N at RGEN 3.3 Ohm, 84 ns P at 1 Ohm, longer on EFM8 port drive), then A\_X\_5\_96 (Matrix II) after the V2 shoot-through check (PINMAP 7.3); the DS Qg of 23 / 33 nC disagrees 3-5x with Ciss 950 pF, so V2 measures the gate edges.

# OpenDrone

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File: esc\_channel.kicad\_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

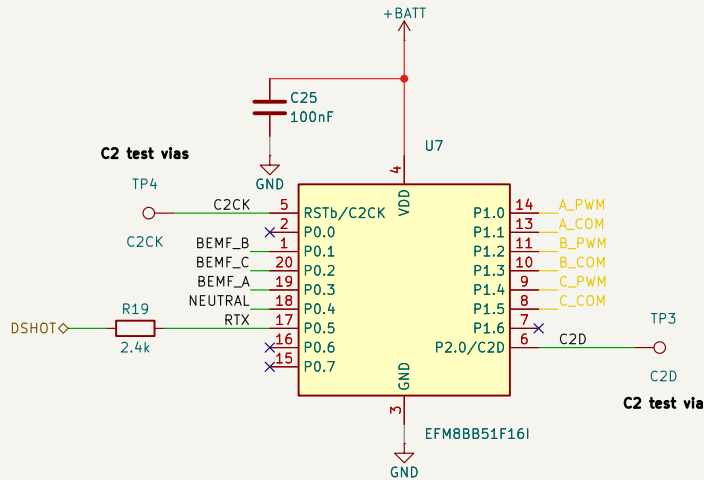
Size: A4 Date: 2026-10-09

KiCad E.D.A. 10.0.6

Rev: rev1

Id: 3/14

## ESC MCU: EFM8BB51, Bluejay BB51 layout A



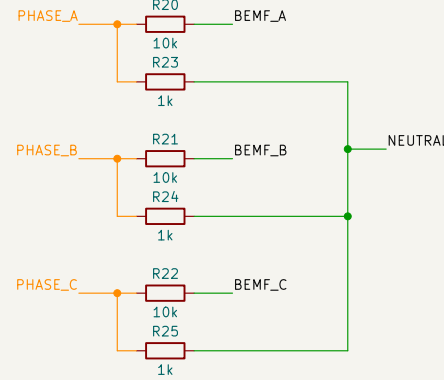
Pin use = Bluejay src/Layouts/BB51/A.inc (0368d11): P0.1 B\_Mux, P0.2 C\_Mux, P0.3 A\_Mux, P0.4 V\_Mux (neutral), P0.5 RTX; P1.0/P1.1 A\_Pwm/A\_Com, P1.2/P1.3 B, P1.4/P1.5 C (PWM active high = N gate, COM active low = P gate, all push-pull); P2.0 DebugPin = C2D. Pin numbers: EFM8BB51 data sheet Rev 1.0 Table 6.1. P0.0, P0.6, P0.7, P1.6 unused: no-connect (PINMAP 7.1).

VDD pin 4 straight on +BATT, no series R (it must track the P sources); C19 100 nF X6S at pin 4 / GND pin 3. The data sheet's 1 uF (Fig. 5.1) is the phase B hot-loop cap C23 10 uF X6S (about 3-4 uF effective at 4.2 V); BR-05 as amended (P4 critique): within 4 mm of pin 4 through the +BATT plane, 100 nF at the pin (v3 cell site 3.4-3.9 mm). ESC3 has no extra cap (C30 deleted, D77). VDD abs max 5.5 V vs 4.35 V cell + ripple + regen 4.7-5.25 V (V3c).

DShot (P4-8): R12 2.4k at RTX. At a 2.5 V cell the 3.3 V DShot high sits up to 0.53 V over P0.5's VDD + 0.3 V abs max; R12 holds the clamp current to  $\leq 0.35 \text{ mA}$   $((3.33 - 2.5) \text{ V} / 2.4 \text{ k})$ ; about 0.1-0.2 mA after the clamp drop) against 50 mA per pin. RSTb/C2CK: internal pull-up only (PINMAP F12).

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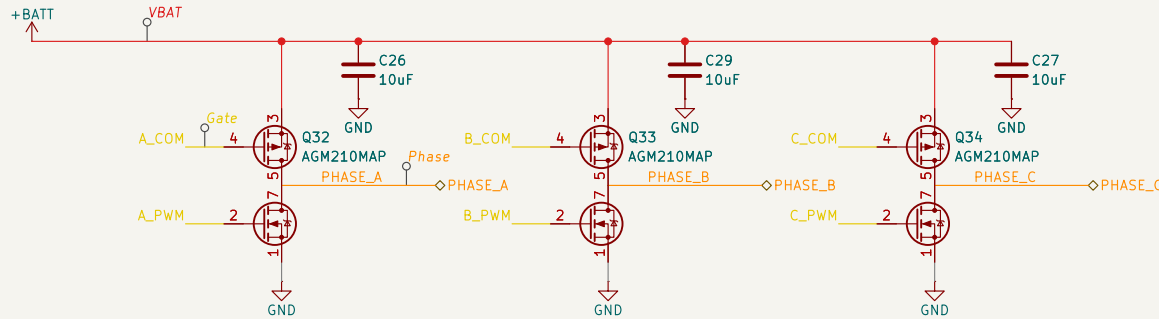
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# OpenDrone

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File: esc\_channel.kicad\_sch

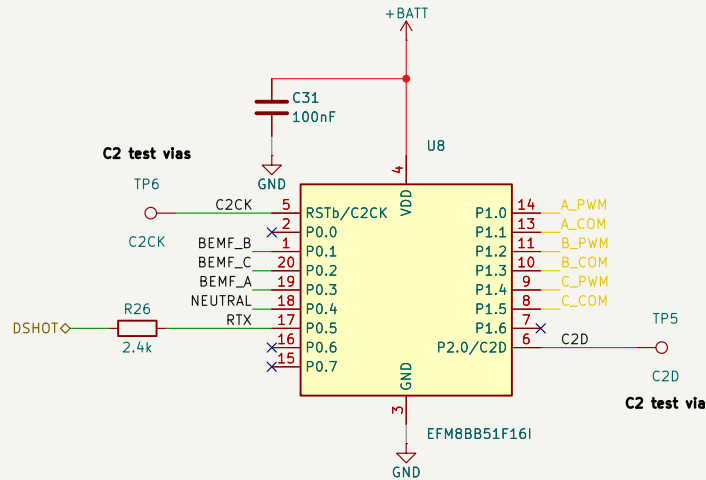
**Title: ESC channel: EFM8BB51 + 3x AGM210MAP**

Size: A4 Date: 2026-10-09  
KiCad E.D.A. 10.0.6

Rev: rev1  
Id: 4/14



## ESC MCU: EFM8BB51, Bluejay BB51 layout A



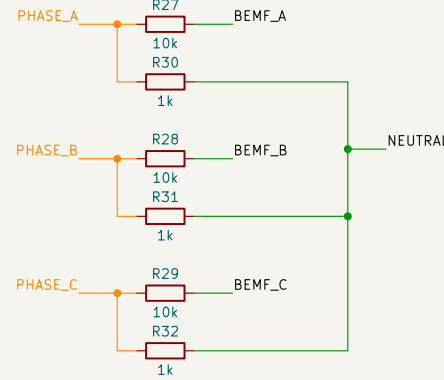
Pin use = Bluejay src/Layouts/BB51/A.inc (0368d11): P0.1 B\_Mux, P0.2 C\_Mux, P0.3 A\_Mux, P0.4 V\_Mux (neutral), P0.5 RTX; P1.0/P1.1 A\_Pwm/A\_Com, P1.2/P1.3 B, P1.4/P1.5 C (PWM active high = N gate, COM active low = P gate, all push-pull); P2.0 DebugPin = C2D. Pin numbers: EFM8BB51 data sheet Rev 1.0 Table 6.1. P0.0, P0.6, P0.7, P1.6 unused: no-connect (PINMAP 7.1).

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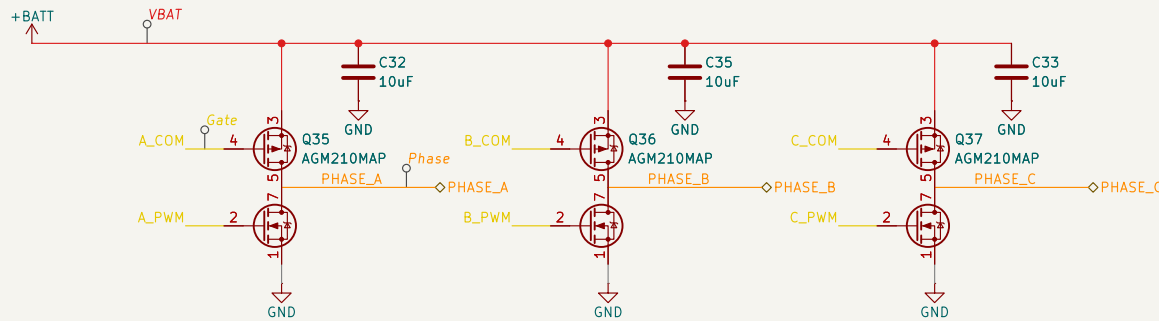
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USB only: the PIO DShot idle-high back-feeds +BATT through the EFM8 I/O diode,  $\leq (3.3 \text{ V} - \text{VF}) / 2.4\text{k}$ , about 1 mA per channel; the EFM8s may cycle near POR, +5V stays on USB via the TPS2116 (10k fallback if V9 shows brown-out cycling).

## Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP (AGMSEMI, DS VER2.73 p1 top view: D23 rule, P3 symbol/footprint check): 1 S1 N source = GND, 2 G1 N gate = x\_PWM, 3 S2 P source = +BATT, 4 G2 P gate = x\_COM, 5/6 + EP D2 (P drain) and 7/8 + EP D1 (N drain) = PHASE\_x; the two drain pads are joined only by the phase island on the board (never split it). Gates straight from the EFM8: no gate resistors and no OR options (tinyPEPPER / Matrix II field record; bom\_plan v3 has none); the about 0.15 A gate-edge current for tens of ns is a transient above the EFM8's 50 mA per-pin abs max, accepted as on the Matrix II (spec 4.4). VGS  $\pm 12 \text{ V}$ , VDS 20 V abs max vs VDD  $\leq 5.25 \text{ V}$  regen peak. Hot loop (D77, P4 critique): one cap per AGM210MAP, C20 / C23 / C21 10 uF X6S 0402 from pin 3 S2 (+BATT) to pin 1 S1 (GND) at the cell's lead side, both pads via-in-pad, so the commutation loop closes at the pins (Matrix II: one cap per FET); the three are also the ESC +BATT bulk. Bluejay: first power-up A\_X\_10\_96 (about 200 ns; DS td(off) + tf = 58 ns N at RGEN 3.3 Ohm, 84 ns P at 1 Ohm, longer on EFM8 port drive), then A\_X\_5\_96 (Matrix II) after the V2 shoot-through check (PINMAP 7.3); the DS Qg of 23 / 33 nC disagrees 3-5x with Ciss 950 pF, so V2 measures the gate edges.

# OpenDrone

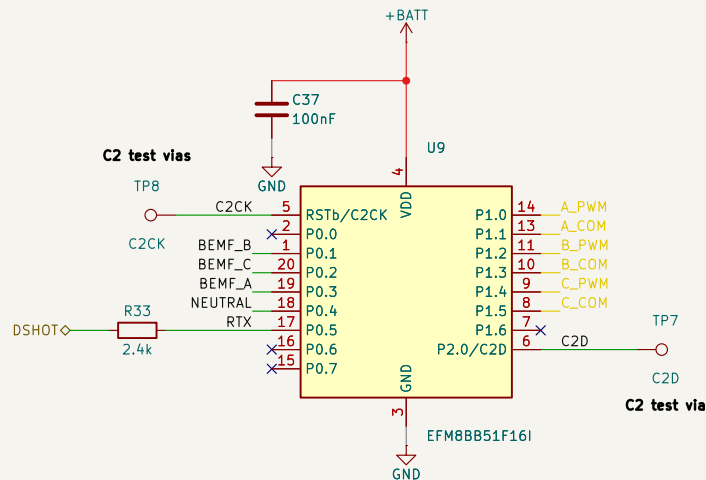
Sheet: /ESC3/  
File: esc\_channel.kicad\_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

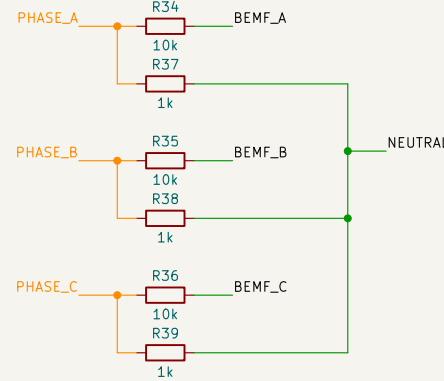
Size: A4  
KiCad E.D.A. 10.0.6

Date: 2026-10-09  
Rev: rev1  
Id: 5/14

## ESC MCU: EFM8BB51, Bluejay BB51 layout A



## BEMF network: 10k series + 1k neutral star



Source: tinyPEPPER single\_esc.sch (CERN-OHL v1.2, wire trace in research 08): 10k series from each phase to its comparator input (BEMF\_A P0.3, BEMF\_B P0.1, BEMF\_C P0.2) and 1k from each phase to the virtual neutral (P0.4 V\_Mux, CMP0 negative). No leg to GND: at 1S the phase stays within the cell + one body-diode drop, and the 10k limit the input clamp current to  $< 0.1 \text{ mA}$ , 1k star, worst case one phase low and two high:  $2/3 \times 5.25 \text{ V} / 1\text{k} = 3.5 \text{ mA}$ , 12 mW per 0201 (50 mW part). P4-B: the netlist puts BEMF\_A/B/C and NEUTRAL on U6 pins 19/1/20 and 18 = A.inc A\_Mux 3 / B\_Mux 1 / C\_Mux 2 / V\_Mux 4.

ESC channel, one file placed 4x. The references drawn are ESC1's; the root sheet offsets them per instance (ESC2 U+1 Q+3 R+7 C+6 TP+2, ESC3 x2, ESC4 x3), matching bom\_plan.json:

ESC1 M1 rear right: U6 C19, hot loop C20 C23 C21, R12-R18, Q29-Q31, TP1/TP2

ESC2 M2 front right: U7 C25, hot loop C26 C29 C27, R19-R25, Q32-Q34, TP3/TP4

ESC3 M3 rear left: U8 C31, hot loop C32 C35 C33, R26-R32, Q35-Q37, TP5/TP6

ESC4 M4 front left: U9 C37, hot loop C38 C41 C39, R33-R39, Q38-Q40, TP7/TP8

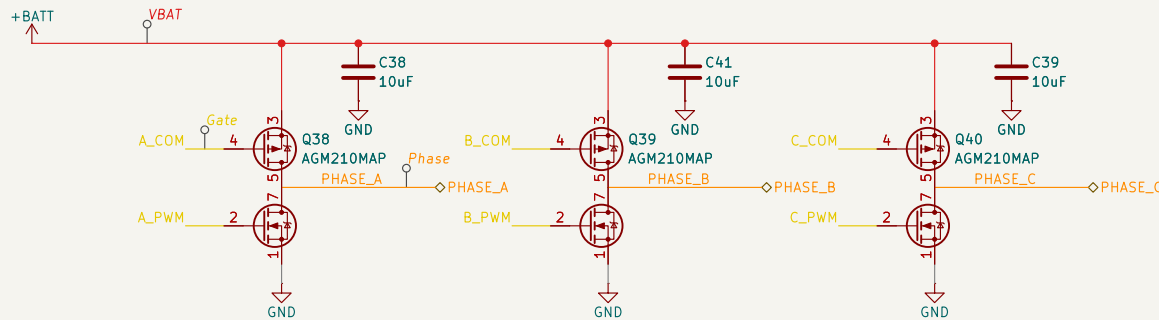
Pins: DSHOT (left, root net /MOTORn: FC GPIO25/24/23/22); PHASE\_A/B/C (right, local label + hierarchical label, net /ESCn/PHASE\_x, motor lands J4/J7/J10/J13). In-sheet nets: x\_COM, x\_PWM (Gate class), BEMF\_x, NEUTRAL, RTX, C2D, C2CK. Netclass flags: Phase (PHASE\_A), Gate (A\_COM), VBAT (+BATT rail): setup\_board.py patterns /ESC?/PHASE\_?, /ESC?/?\_COM, /ESC?/?\_PWM assign the rest.

Source: copied from OpenAIO hardware/esc\_channel.kicad\_sch (frames, block titles, C2 test points at the MCU pins, three-column BEMF block, power stage along the bottom) and adapted: AT32F421 + NSG2065Q + 6x DOY180N03T become EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, Matrix 1S 5IN1 II stage, D56), the 3S divider becomes the tinyPEPPER 1S star.

Power-up (P4-11, PINMAP F16): battery first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, tPOR  $\leq 390 \text{ us}$ ) and the Bluejay bootloader turns on the 70-220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1-2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV full), through  $\geq 72\text{k}$  (pull-up + R12), i.e.  $\leq 60 \text{ uA}$ ; 0.72 V over the 3.63 V unpowered-FT limit, current-limited. Accepted on Matrix parity (G473 FT pads about 4 V unpowered); V9 scopes the pad at power-up; fallback a 3.3 V-domain clamp or a DShot-path enable (not in rev1).

USB only: the PIO DShot idle-high back-feeds +BATT through the EFM8 I/O diode,  $\leq (3.3 \text{ V} - \text{VF}) / 2.4\text{k}$ , about 1 mA per channel; the EFM8s may cycle near POR, +5V stays on USB via the TPS2116 (10k fallback if V9 shows brown-out cycling).

## Half-bridges A/B/C: AGM210MAP P+N, one package per phase



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# OpenDrone

Sheet: /ESC4/

File: esc\_channel.kicad\_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

Size: A4

Date: 2026-10-09

Rev: rev1

KiCad E.D.A. 10.0.6

Id: 6/14

decoupling caps. place close to IC, one is enough when 2 pins are next to each other

**VBAT divider**

**current sense RC, 1.6 kHz**

### Circuit diagram

The diagram illustrates the boot strap and SVD test vias for the XPR2350. It shows the connection of the boot strap (R44) and the SVD test vias (TP9 FCB, TP12 CLK, TP13 DIO) to the QSPILSS, SWCLK, and SWDIO pins respectively. The boot strap is connected to the XPR2350 pin 60 (BOOT) and the FC\_BOOT pin. The SVD test vias are connected to the XPR2350 pins 12 (CLK) and 13 (DIO) respectively. The diagram also shows the connection of the boot strap (R44) and the SVD test vias (TP9 FCB, TP12 CLK, TP13 DIO) to the QSPILSS, SWCLK, and SWDIO pins respectively.

**Crystal per datasheet. 10pF loading = 2x 15pF in series + -2.5pF stray**

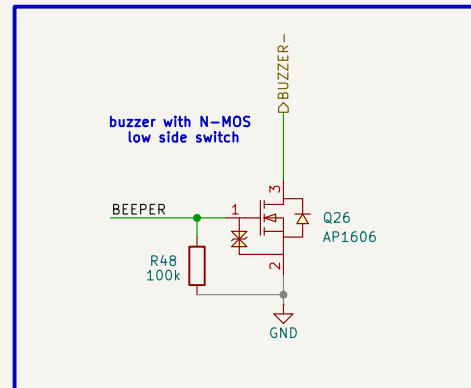
X1, Abracon ABM8-272-T3 = the crystal RP2350 D5 8.21.1 Table 629 names (11 MHz, CL 10 pF, ESR <= 50 Ohm, -40..85 C); C42/C43 15 pF and the XOUT 1k R40 are the RPI quiescent values for it (OpenF5 R5), D78: replaces the 150 Ohm 2520 part. X1, C42, C43, R40 at pins 21/22, nothing switching under them.

**Boot strap + SVD test vias**

R44 1k from QSPILSS (pin 60, BOOT) to FC\_BOOT: the shared boot SWD1 puts FC\_BOOT low through D8 (RX sheet, D65): low at reset = BOOTSEL / Uf2. 1k (RPI hardware guide R6): Csn <= 0.51 V with the diode drop (VIL 0.8 V). Keep the QSPILSS-to-R44 stub <= 1 mm (in-package flash CS, RPI guide R6): any length after R44, TP9 FCB, TP12 CLK (SWCLK), TP13 DIO (SWDIO): mask-open filled 0.35/0.20 via (D67), no silk, out of the BOM and the CPL listed in the pinout doc (the house sheet leaves SWD open). Source: OpenF5-Lite-Mini-RP2350a BOOTSEL strap, button moved to the RX sheet.

[illegible]

+3V3  
 R41 27R  
 C57 4.7uF  
 GND  
 VREG\_AVD0  
 PWR\_FLAG  
 USB\_OTG\_VDD  
 GPIO19 13 CYGRO\_INT  
 GPIO15 14 SPI01\_SCK  
 GPIO11 15 SPI01\_MISO  
 GPIO12 16 SPI01\_MISO  
 GPIO13 17 CYGRO\_CS  
 GPIO18 20 SPI01\_SCK  
 GPIO19 21 SPI01\_MISO  
 GPIO20 22 SPI01\_MISO  
 GPIO21 33 FLASH\_CS  
 GPIO22 34 MOTOR4  
 GPIO23 35 MOTOR3  
 GPIO24 36 MOTOR2  
 GPIO25 37 MOTOR1  
 GPIO19 2 QUART0\_TX  
 GPIO1 3 QUART0\_RX  
 GPIO8 12 LED\_STRIP  
 GPIO17 28 PI4QWART0\_TX  
 GPIO3 5 PI4QWART0\_RX  
 GPIO8 13 QUART1\_RX  
 GPIO4 7 QUART1\_TX  
 GPIO4 9 CYGRO\_CLKIN  
 GPIO7 10 LED0  
 GPIO26 40 LED1  
 GPIO27 41 HD\_EN  
 GPIO28 42 HD\_EN  
 GPIO29 43 HD\_EN  
 GPIO14 18 OSD\_V  
 GPIO15 19 OSD\_EN  
 GPIO16 27 OSD\_SYNC



PIO2 framebuffer OSD: W, EN, SYNC consecutive

P433: All 1<sup>st</sup> power-up and 5 V-tolerant pads (PINMAP 4.5, I/F: RP2320 SS Mode 1433): all 1<sup>st</sup> pad would exceed ~3.63 V while VID0 is 0.8-1.8V @ 2.4 at 2.5 V. 5.5 V @ 2.5 V.

MOTOR1\_4 (GP1025-22\_FET), battery first; each EFM8 weak pull-up (70–220 kOhm to VDD) = cell reaches the pad through its 2.4k before +393 is up to 3.63 V. If the pad were pulled down by 100k, it would reach 3.63 V. The current is circled = 0.72 V over 3.63 V. so ~ 72 kOhm (<= 60 uA). Matrix padding GP1027 F pins -4 w/ VDD IO. v9 spikes it; fallback (P16): 3.3 V domain clamp. No choice.

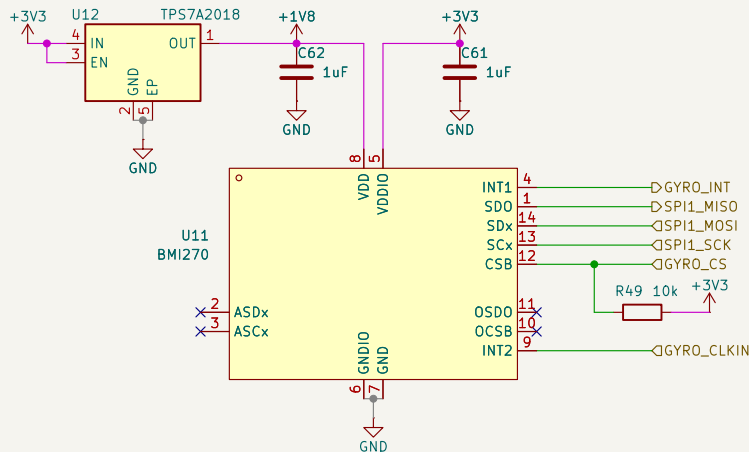
ADC-VBAT (LP129): R45/R46 put up to 2.18 V on the pad from 5 kOhm while VIDAT < 1.7 V, so the pad clamp to IOVDD conducts ≈ 0.3 mA during low VIDAT.

IOVDD (GP1027): R11 2.4 k and the BASE16D from OA\_EN (<= 90 uA) hold it at < 6.22, before driving reset; inside IOVDD + 0.5, and below the VBAT. AP1616H has 100k pull-down to GND. The driver is also on VDD (VOH, 1.4 mA into R11) or Vbat(Ths) max 1.2 V.

AP1616H: 100k pull-down to GND. The driver is also on VDD (+393 V, no excursion, UART1/Y1/RX1); the 0.4 feds only through U5 (needs IDEN, high, none, USB /4-DMA); the host drives only after the device pull-up.

U5: 100k pull-down to GND. The driver is also on VDD (+393 V, no excursion, 3.3 V logic only (README): a 5 V TX would exceed 3.63 V at power-up).

## Universal IMU wiring and LGA-14 land (BMI270, ICM-42688-P)



GYRO BMI270: U11, R49

Source: OpenFC-Lite-Mini imu.kicad\_sch (universal LGA-14 land, SPI1 4-wire, R49), copied and adapted. BMI270 on the BOM; TDK ICM-42688-P qualified on the same land (V10 fly-off, one part per revision).

Pins 2/3 (BMI270 ASDx/ASCx, TDK RESV) and 10/11 (OCSB/OSDO, TDK RESV): no-connect. BMI270 DS Table 22: unused ASDx/ASCx go to VDDIO or stay open, never to GND (the house sheet grounded them for the LSM6DSV16X).

Pins 6/7 GND (TDK pin 7 RESV must be GND). VDDIO pin 5 = +3V3 (RP2354A I/O level). VDD pin 8 = +1V8; the BMI270 has no supply-order rule (DS 4.3). Abs max 4.0 V on VDD/VDDIO, I/O up to VDDIO + 0.3 V.

Pin 9: TDK CLKIN = GYRO\_CLKIN (RP2354A GPIO6, PWM 32 kHz, driven only with the TDK part); BMI270 INT2, unused, left disabled. R49 10k; GPIO13 resets with its pull-down on, which would select the chip; R49 holds CSB >= 2.75 V.

P5: top side next to U12, LGA rotated 90 deg; GYRO\_1\_ALIGN from the placed orientation (PINMAP F11).

+1V8 GYRO LDO: U12, C61, C62 (D52 BR-09, P4-17)

Source: pcb-agent-commons blocks/LDO\_IMU\_TPS7A2033 (OpenFC-H7) topology with the 1.8 V part and no ferrite (fed from the +3V3 LDO): the house OpenFC-Lite-Mini also runs the gyro VDD from its own 1.8 V LDO. TPS7A2018 DQN: 1 OUT, 2 GND, 3 EN, 4 IN, 5 EP to GND. IN = EN = +3V3; VIN 1.6-6.0 V, VEN 0-6.0 V (abs max 6.5 V), EN never above IN. Load < 1 mA.

No 100 nF at the gyro: C61 = U12 CIN at BMI270 VDDIO pin 5, C62 = U12 COUT at VDD pin 8 (COUT >= 0.47 uF effective, ESR <= 100 mOhm; about 0.6-0.85 uF at 1.8 V). C61 counts in the +3V3 COUT sum (P4-12, POWER).

P5: U12 on the gyro side with OUT facing pin 8; C61 between U12 pin 4 and gyro pin 5, C62 <= 1.5 mm from pin 8 (P5-06), else re-place U12.

# OpenDrone

Sheet: /IMU/

File: imu.kicad\_sch

**Title: Gyro BMI270 + 1.8 V LDO**

Size: A5

Date: 2026-10-09

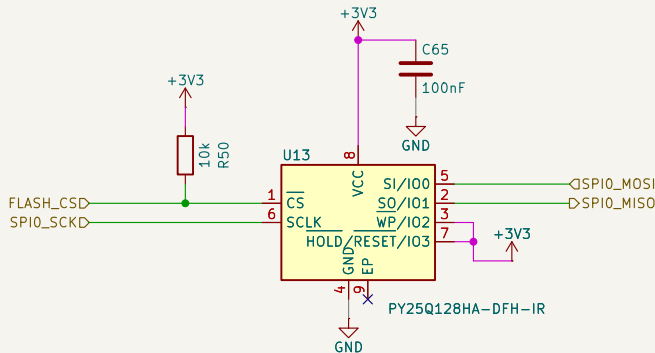
Rev: rev1

KiCad E.D.A. 10.0.6

Id: 8/14



## SPI NOR PY25Q128HA: 16 MB blackbox on SPI0 (U13 C65 R50)



Placement (P5): beside the RP2354A on SPI0 (bom\_plan region nor, top; either side works). U13, SPI0 and FLASH\_CS stay  $\geq 3$  mm from the RX antenna hole (spec 4.9 and 11, DRU 'SPI0 3 mm from the RX antenna hole'): SPI0 harmonics fall in the ELRS band, V7 measures desense while logging. C65 at pin 8. Filled+capped 0.35/0.20 via-in-pad allowed in the lead lands, never in the tabs. RP2354A pads reset to 4 mA drive and slow slew (EMI lever without parts). Sourcing: no LCSC listing; NextPCB buys by MPN (D60), the quote confirms it. Same-pinout fallbacks in the Betaflight table (USON-8 4x4 land not staged): PY25Q128HA-QVH-IR, GD25Q128EQIGR.

Source: OpenFC-Lite-Mini hardware/blackbox.kicad\_sch (rev2). Kept: layout, the 10k CS pull-up (R39  $\rightarrow$  R50) and the 100n supply cap (C43  $\rightarrow$  C65). Changed: its microSD (Card1) became this 16 MB SPI NOR (D66); the SPI0(SCK,MOSI,MISO) bus label became four hierarchical labels (sch\_contract.json).

U13 Puya PY25Q128HA-DFH-IR: 128 Mbit, USON-8 3x3 (DF), 2.7-3.6 V, -40..85 C, fC 133 MHz, READ 03h 80 MHz, JEDEC ID 85 20 18 is in the Betaflight m25p16 table: USE\_FLASH\_M25P16, SPI0 PA18/PA19/PA20, FLASH\_CS\_PIN PA21 (PINMAP). SPI0 writes at 75 MHz and reads at 50 MHz: inside fC and fR.

WP# (3) and HOLD#/RESET# (7) tied to +3V3: the DS gives an internal pull-up only to the SOP-16 dedicated RESET#, so pin 7 must not float. Delivered SR = 00h, CR = 00h (QE 0, HOLD# function, WPS 0: no block lock) and Betaflight sends 1-bit SPI commands only, so IO2/IO3 never drive against the rail (the ERC pin\_to\_pin warning on pins 3/7 against a +3V3 power output is this deliberate tie).

R50 10k: CS# must follow VCC at power-up (DS 5.5). RP2354A pads reset with the pull-down on (PDE 1, RPD 36-113 kOhm at 3.3 V): CS >= 36/46 VCC = 0.78 VCC > VIH 0.7 VCC, so U13 stays deselected through reset and boot. U13 and the RP2354A IOVDD share +3V3: no pin above VCC + 0.5 V.

C65 100 nF X6S at pin 8, GND return straight to pin 4 (ICC max 13 mA read, 12 mA erase; adds 0.1 uF nominal to the +3V3 COUT sum of P4-12). EP (9) = two tie-bar tabs with no electrical function in the DS: no-connect, no vias in the tabs.

# OpenDrone

Sheet: /BLACKBOX/

File: blackbox.kicad\_sch

**Title: Blackbox 16 MB SPI NOR PY25Q128HA**

Size: A5

Date: 2026-10-09

Rev: rev1

KiCad E.D.A. 10.0.6

Id: 9/14

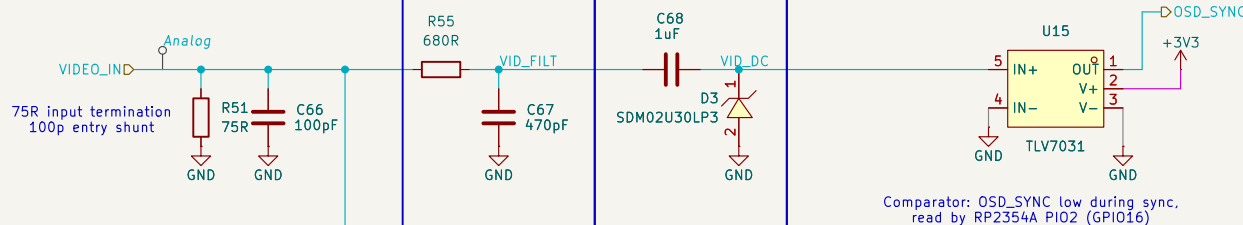
# VIDEO PATH

Chroma Filter  
fc = 500 kHz

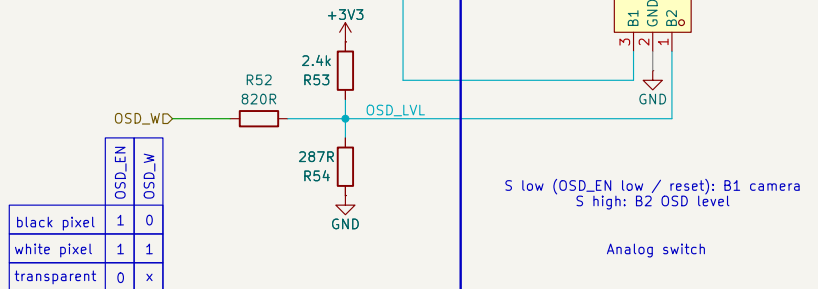
DC Restoration

Sync detection, VID\_DC sync tips  
clamped to  $-VF$  (about  $-0.1\text{--}0.2\text{ V}$ )

U15 V+ (2) is decoupled by C70 in the switch cell,  
one 100n shared with U14 (BR-13)



Resistor divider lets the OSD\_W GPIO select  
white (high: 1.05 V) or black (low: 0.27 V)



S low (OSD\_EN low / reset): B1 camera  
S high: B2 OSD level

Analog switch

VIDEO\_OUT: the switch output drives the RTC6705 video network on the VTX sheet directly (R58 1.2k || C104 33p, R59 270R, C105 4.7u, R60 1.2k: OpenOSD-X values); the house COS8051 buffer (U1, R46-R48, C41) is deleted (spec 4.8). OSD levels (P4-10, recomputed at the P4 critique with the real load): open-circuit at OSD\_LVL (+3V3 3.30 V, OSD\_W push-pull) black 0.268 V / white 1.054 V from 195R (R53) | R54 = 0.352 V / 256R, with R52) + switch on 7-18R. VIDEO\_OUT is loaded by the VTX network ahead of C105: R58 + R59 = 1.47 kOhm at DC, about 0.27 kOhm above 5 MHz (C104 bypasses R58). Loaded levels: black 0.234-0.236 V, white 0.920-0.927 V (swing x0.88 of the open-circuit 0.786 V); edges above 5 MHz about x0.57 vs about x0.85 for the camera (37.5R source): OSD text slightly dimmer and softer than the camera picture, not a failure (the house COS8051 buffer that made the source low-impedance is deleted). V9 checks OSD white / black and edge sharpness on a monitor; fallback: rescale R52-R54 within the 3 mA OSD\_W budget. PS-01 (R54 270R) would lower white to about 0.89 V: not recommended. OSD\_W: 2.74 mA high ( $\leq 3\text{ mA}$ , 2.79 mA at 3.36 V), 0.33 mA low.

Source: OpenFC-Lite-Mini hardware/osd.kicad\_sch (rev2, flown), copied and adapted: cells, texts, truth table and part spots kept (A3 -> A4). Same topology: 75R termination, 680R/470p chroma filter, 1u + SDM02U30LP3 sync-tip clamp, TLV7031 comparator, SN74LVC1G3157 camera/level switch (contract source, spec 4.8).

Changed: COS8051 buffer deleted (output = VIDEO\_OUT); divider R42 3.9k / R43 12k / R44 1.4k -> R52 820R / R53 2.4k / R54 287R (about 5x down, P4-10); C39 + C42 -> one shared C70 100n X65 (D52 BR-13 + BR-18); C66 100p entry shunt added (spec 11); refs per bom\_plan (R41>R51, C38>C67, C40>C68, D9>D3, U2>U15, R45>R55, R42-44>R52-54, U18>U14). Symbols from the OpenDrone catalogue with TI pin numbers: TLV7031 X2SON 1 OUT, 2 V+, 3 V- (centre pad), 4 IN-, 5 IN+, SN74LVC1G3157 DTB 1 B2, 2 GND, 3 B1, 4 A, 5 VCC, 6 S. The house lib symbols used the EasyEDA pad order (consistent with its own footprints), so the drawing is re-routed with the same nets: IN+ = VID\_DC, IN- = GND, B1 = camera, B2 = OSD level. The house sheet's rotated ground is upright here.

Operation: OSD\_EN low (RP2354A reset pull-down, OSD idle) -> B1, the camera passes through. OSD\_EN high -> B2: OSD\_W high = white, low = black. No OSD without camera sync (osd\_pico.c overlays on detected sync). OSD\_W/EN/SYNC = GPIO14/15/16, consecutive, PIO2 (PINMAP).

Sync threshold (P4-10): 680R/470p = 498 kHz: colour burst x0.11-0.14 (+-17-21 mV), 4.7 us sync kept (tau 0.32 us). C68 + D3 (cathode on VID\_DC) clamp the sync tips to  $-VF$ ; the leakage-limited clamp current (uA average, tens of uA at the tip) gives VF about 0.1-0.2 V over temperature (I: DS 0.37 V max at 10 mA / 25 C, 0.20 V at 10 mA / 125 C). IN- at GND puts the trip point VF above the tip, 33-67 % of the 0.3 V sync,  $\geq 0.08\text{ V}$  from tip and blanking after VIO +8 mV and VHYS 2-17 mV. OSD\_SYNC is low during sync (osd\_tx.pio waits for a low sync pin). IN+ at  $-VF$  stays inside abs max (VEE - 0.3 V), at most 0.1 V under the ROC input floor (VEE - 0.1 V), no phase reversal (DS 6.4.1). tpd 3 us on both edges (equal shift, as on the house board).

Ratings: +3V3 only (U14 1.65-5.5 V, U15 1.6-6.5 V). S: VIH 0.7 VCC = 2.31 V, edges  $\leq 10\text{ ns/V}$  (fast slew pads). AC-coupled camera sync tips (-0.3 to -0.4 V) are below the switch ROC (0.VCC) but inside abs max (-0.5 V, 50 mA; 37.5R source), as on the house board: V9 checks. Supply: U15 0.9 uA, U14  $\leq 10\text{ uA}$ , divider 0.94-1.26 mA + OSD\_W 2.74 mA. C68 X5R carries  $\leq 0.6\text{ V DC}$ : no DC-bias loss (P4-13 hot-zone rule void, D69).

Placement (P5; P4-17 sheet note): bottom, region osd. BR-13: C70 is the only supply cap of U14 (VCC 5) and U15 (V+ 2): place it between the two parts (2.5 mm apart). GND via-In-pad; the P5 check lists its distance to both pins. U15 centre pad (3, V-) to GND through a via (TI 7.4). R51 and C66 at the video entry on the CAM pad J23 / plug J32 line: camera return into L2 at the CAM GND pad; video on L1/L6 over a plane with GND both sides, never L3 (spec 11). R55, C67, C68, D3 next to U15 IN+. Video parts away from the boost and its inductor (preference, D69; v2 floorplan 3.3-8.6 mm; V5 video test).

Netclass Analog: VIDEO\_IN (flag here), VIDEO\_OUT, VID\_FILT, VID\_DC, OSD\_LVL, OSD\_SYNC by the setup\_board.py patterns: OSD\_W, OSD\_EN Default. Sourcing: exact MPNs from bom\_plan: LCSC empty where bom\_plan has none (C66, R52, R54; open item A9-LCSC). PS-01 (R54 -> 270R RC0201FR-07270RL, shared with R59, -1 line): black 0.258 V, white 1.011 V, 187R, OSD\_W 2.79 mA, also passes P4-10; proposed in contract-issues, drawn = bom\_plan.

# OpenDrone

Sheet: /OSD/  
File: osd.kicad\_sch

Title: PIO analog OSD front end

Size: A4 Date: 2026-10-09

KiCad E.D.A. 10.0.6

Rev: rev1

Id: 10/14

# SX1281 RADIO + 2.4 GHz FILTER + WIRE ANTENNA

**SX1281 RADIO + 2.4 GHz FILTER + WIRE ANTENNA**

**U18 SX1281**

**FL1 2450FM07D0034T**

**AE1 ANT 2.4G wire**

**RF10 (22) -> RF\_RFIO**  
 -> FL1 Johanson  
 2450FM07D0034T (house  
 part: input matched to  
 the SX1281 RFIO,  
 internally DC blocked, IN  
 pin 1 toward the chip)  
 -> RF\_RX\_ANT (50 Ohm)  
 -> AE1: insulated  
 lambda/4 wire (about  
 28-29 mm, trimmed in  
 V7) in a plated D0.5  
 hole; AE1 DNP, out of  
 BOM / CPL. RF class:  
 board rule areas  
 RF\_RX\_ANT /  
 RF\_RX\_EXIT.

**SX1281 (U18), copied from OpenRX-Lite-UFL; SX1281IMLRT (D52 BR-25), SX1280IMLRT drop-in alternate (ranging engine only).**  
**LDO MODE** (Semtech DS Rev 3.2 5.1, Fig. 15-4): DCC\_SW (14) open, the sibling's 15 uH L4 deleted (the ELRS layout has no radio\_dcdc); VDD\_IN (2) tied to DCC\_FB (12) = VREG\_RADIO with C87 470 nF + C88 10 nF; VR\_PA (1) C86 10 nF; VBAT\_IO (11) C91 and VBAT (15) C89 100 nF (Table 15-1) + C92 1 uF radio local (P4-12 lever PS-06); the house sheet's extra 10 nF / 10 uF deleted (BR-06).  
**PIN 5 IS GND** (DS pin table) and is grounded here; the sibling sheets left it open (only XTB stays open with a TCXO, DS 15.2). DI02 / DI03 open.  
**52 MHz TCXO X3** (kept, D53): clipped sine AC-coupled by C84 1 nF into XTA (4), C85 100 nF at VCC, XTB (6) open.

SX1281 (U18), copied from OpenRX-Lite-UFL; SX1281MLTR (D52 BR-25), SX1280MLTR drop-in alternate (ranging engine only).

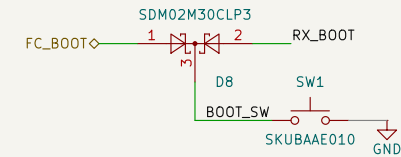
LDO MODE (Sembles DS Rev 3.2 5.1, Fig. 15-4): DCC-SW (14) open, the sibling's 15 uH deleted (the ELRS layout has 10 uH), VDDIO1 (12) tied to DCC-FB (12), VREG-RADIO with C87 470 nF + C88 10 nF, VR-PA (1) C86 10 nF, VBAT-I/O (11) C91 and VBAT (15) C89 100 nF (Table 15-1) + C92 1 uF radio local (PA=12 levered DS-06); the house sheet's extra 10 nF / 10 uF deleted (BR-06).

PIN 5 IS GND (pin5 pin table) and is grounded here; the sibling sheets left it open (only XTB stays open with a 100 nF DS-15.2). D1P / P105 open.

52 MHz XTAL X3 (kept, D53): clipped sine AC-coupled by C84 1 nF into XTA (4), C85 100 nF at VCC, XTB (6) open.

```
RFIO (22) -> RF_RFIO
-> FL1 Johanson
2450FM07D0034T (house
part: input matched to
the SX1281 RFIO,
internally DC blocked, IN
pin 1 toward the chip)
-> RF_RX_ANT (50 Ohm)
-> AE1: insulated
lambda/4 wire (about
28-29 mm, trimmed in
V7) in a plated D0.5
hole; AE1 DNP. out of
BOM / CPL. RF class;
board rule areas
RF_RX_ANT /
RF_RX_EXIT.
```

# BOOT / BIND BUTTON (D65)



## NOTES

RX SHEET (P4 shard C). Sources: the SX1281 radio, 52 MHz TCXO, RGB LED and Wi-Fi pi match are COPIED from the house sheet OpenRX-Lite-UFL hardware/esp32c3\_sx1281\_lite.kicad\_sch (same as OpenAIO rx\_esp32c3\_sx1281.kicad\_sch): same symbol positions and wires, adapted to bom\_plan refs/values/MPNs and to D52/D53/D58 (LDO mode, pin 5 to GND, UFL replaced by the wire-antenna hole, 2450AT07A0100001T chip antenna, grounds turned down); the house Johanson 2450FM07D0034T front-end filter stays. The ESP32-PICO-V3 block is new, from the Espressif ESP32-PICO series datasheet v1.3 Fig. 11 (D52 BR-01, D55); the shared boot / bind button is new (D65, Matrix II style). Pin plan: research/PINMAP.md 6. Parts: hardware/bom\_plan.json. Radio and LED +3V3 come from the POWER sheet (U4 LP5912-3.3).

BOOT / BIND BUTTON (D65; new). SW1 Alps SKUBAAE010 (2.4 x 1.4 mm) from BOOT\_SW to GND. D8 Diodes SDM02M30CLP3 dual Schottky, common cathode on BOOT\_SW: A1 = FC\_BOOT (RP2354A QSPILSS through R44 1k, RP2350A sheet), A2 = RX\_BOOT (ESP32 GPIO0, pin 23). Released: each strap sees only diode leakage (IR <= 0.15 uA at 10 V, 25 C), so GPIO0 keeps its internal pull-up and QSPILSS its own level. Held while power is applied: RP2354A BOOTSEL and ESP32 ROM download mode (CSn <= 0.51 V vs VIL 0.8 V; GPIO0 <= 0.42 V vs VIL 0.825 V); each mode stays latched until that chip's next reset. Production / ELRS recovery (D67, spec 12.1): (1) hold SW1 (or probe BOOT\_SW) while powering; (2) write the Betaflight UF2 (or picotool reboot); the RP2354A restarts into Betaflight, the ESP32 stays in download mode; (3) esptool through Betaflight serial passthrough. Pressed while running: ELRS button (bind 1.5 s, Wi-Fi 5 s, reboot 12 s); 2.9 mA through R44, CS stays >= 2.62 V, the FC keeps running. Bench V-btn: if GPIO0 dips below 2.5 V under XIP load while pressed, add 100 pF 0201 on BOOT\_SW.

ESP32-PICO-V3 (U16), Espressif PICO datasheet v1.3 Fig. 11 / Table 4 (new block). SUPPLY: VDDA (1, 43, 46), VDDA3P3 (3, 4), VDD3P3\_RTC (19), VDD3P3\_CPU (37) on +3V3; C82 10 uF + C76 100 nF at VDDA3P3 pins 3/4 (Fig. 11 C1/C2). The 4 MB flash, 40 MHz crystal, CAP1/CAP2 network and the VDD\_SDIO capacitor are inside the SIP (Fig. 8); VDD\_SDIO (26) stays open. EN: CHIP\_PU (9) = RX\_EN, R56 10k to +3V3 + C73 1 uF (tau 10 ms), never floating. STRAPS at reset (P4-6, checked on the netlist): GPIO0 (23) RX\_BOOT: internal 45k pull-up, D8 blocks -> 1 = SPI boot (SW1 held -> download). GPIO2 (22) PA\_EN: internal 45k pull-down + LP5907 EN 1 MOhm pull-down (VTX sheet), no external part -> 0. GPIO5 (34) VTX\_SPL\_CLK: internal pull-up, R81 1k into the RTC6705 CMOS input -> 1 (default timing). GPIO12 / MTDI (18): no connect, internal pull-down -> VDD\_SDIO 3.3 V for the in-package flash (VDD\_SDIO eFuse burn optional). GPIO15 / MTDI (21): no connect, internal pull-up -> boot log on U0TXD (FC ignores it). NOT CONNECTED: flash CMD/I011 (30) and CLK/I06 (31); NC pins 25, 35, 36, 44, 45, 47, 48; free I07/8/9/10 (32, 33, 28, 29; PSRAM on the -02 alternate), I020 (27), I35 (11), I38 (7), I39 (8). ELRS LAYOUT JSON (board overlay, D53): serial\_Lrx 3, serial\_Ltx 1, radio\_sck 25, radio\_mosi 32, radio\_miso 33, radio\_nss 27, radio\_rst 26, radio\_busy 36, radio\_dio1 37, led\_rgb 22, button 0, vtx\_nss 19, vtx\_sck 33, vtx\_mosi 14, vtx\_miso unset, vtx\_amp\_pwm 13, vtx\_amp\_vpd 4, vtx\_amp\_vref 2; patched ELRS: GPIO21 VTX\_PWR\_CTL (+3V3-VTX LDO EN through R84 1k), GPIO34 PA\_NTC (ADC1, read in the iStick window, erratum 3.11). VTX LINES: series 1k R80/R81/R82 are drawn on the VTX sheet and placed at the ESP32 end (back-feed limit with +3V3-VTX off, F21). GPIO13 VTX\_AMP\_PWM = ELRS vtx\_amp\_pwm (RTC6705 PAOUT1 feed PNP, not the SE5004L bias). PA\_DET is ADC2\_CH0 (not readable with Wi-Fi on; not a flight state). +3V3 LOAD: ELRS about 0.07 A in flight. Wi-Fi TX peaks 370 mA (DS Table 19, 802.11b 19.5 dBm) in ELRS Wi-Fi mode only (disarmed, bench); +3V3 peak about 0.48 A with the FC and LED, inside the LP5912 500 mA rating; LDO heat and the COUT sum belong to the POWER sheet (P4-12). PLACEMENT: SIP site from floorplan v3 (D55, D61); no 85 C halo rule (D69). Alternate part ESP32-PICO-V3-02 (pins 28/29 unused here, P4-14).

BOOT / BIND OUTMODE (D65; new), SW1 Alps SKUBAAE010 (2.4 x 1.4 mm) from BOOT\_SW to GND, D8 Diodes SMD2M30CLP3 dual Schottky, common cathode on BOOT\_SW: A1 = FC\_BOOT (RP2354A QSPI\_SS through R44 1k, RP2350A sheet), A2 = RX\_BOOT (ESP32 GPIO0, pin 23). Released; each strap sees only diode leakage (IR <= 0.15 uA at 10 V, 25 C), so GPIO0 keeps its internal pull-up and QSPI\_SS its own level.

Held while power is applied: RP2354A BOOTSEL and ESP32 ROM download mode (CSn <= 0.51 V vs VIL 0.8 V, GPIO0 <= 0.42 V vs VIL 0.825 V); each mode stays latched until bit chip's next reset. Prior to boot, ELRS binary (under the supply) (1) holds CSn (4) held low (or probe BOOT\_SW while pressing); (2) write the Betaflight UFL (or pitocout reboot); the RP2354A restarts into Betaflight, the ESP32 stays in download mode; (3) espboot through Betaflight serial pass through.

Pressed while running: ELRS button (bind 1.5 s, Wi-Fi 5 s, reboot 12 s); 2.9 mA through R44, CS stays >= 2.62 V, the FC keeps running.

Bench V-btn: if GPIO0 dips below 2.5 V under XIP load while pressed, add 100 pF 0201 on BOOT\_SW.

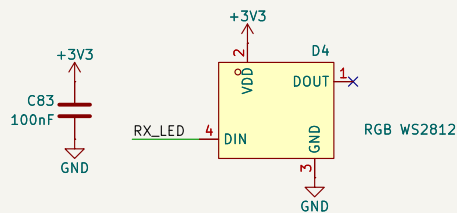
```

ESP32-PICO-V3 (U16), see Pico PICO datasheet v1.3 (Fig. 11 / Table 4 (new block)).
SUPPLY: VDDA (1, 43, 46), VDDA3P3 (3, 4), VDD3P3_RTC (19), VDD3P3_CPU (37) on +3V3; C82 10 uF
+ C76 100 nF at VDDA3P3 pins 3/4 (Fig. 11 C1/C2). The 4 MB flash, 40 MHz crystal, CAP1/CAP2
network and the VDD_SDIO capacitor are inside the SiP (Fig. 8): VDD_SDIO (26) stays open.
EN: CHIP_PU (9) = RX_EN, R56 10k to +3V3 + C73 1 uF (tau 10 ms), never floating.
STRAPS at reset (P4-6, checked on the netlist):
  GPIO0 (23) RX_BOOT: internal 45k pull-up, D8 blocks -> 1 = SPI boot (SW1 held -> download),
  GPIO2 (22) PA_EN: internal 45k pull-down + LP5907 EN 1 Mohm pull-down (VTX sheet), no external
  part -> 0.
  GPIO5 (34) VTX_SPL_CLK: internal pull-up, R81 1k into the RTC6705 CMOS input -> 1 (default
  timing).
  GPIO1 / MTDI (18) no connect, internal pull-down -> VDD_SDIO 3.3 V for the in-package flash
  (VDD_SDIO eFuse burn optional).
  GPIO15 / MTDI (21): no connect, internal pull-up -> boot log on U0TXD (FC ignores it).
NOT CONNECTED: flash CMD/I011 (30) and CLK/I06 (31); NC pins 25, 35, 36, 44, 45, 47, 48; free
I07/I8/I9 (32, 33, 28, 29; PSRAM on the -02 alternate), I020 (27), I35 (11), I38 (7), I39 (8).
ELRS LAYOUT JSON (board overlay, D53): serialLrx 3, serialTx 1, radio_ssc 25, radio_omiso 32,
radio_miso 33, radio_nss 27, radio_rst 26, radio_busy 36, radio_dio1 37, led_rgb 22, button 0, vtx_nss 19,
vtx_ssc 5, vtx_omiso 14, vtx_omiso unset, vtx_omiso_pwm 13, vtx_omiso_vpd 4, vtx_omiso_vref 2; patched
ELRS: GPIO121 VTX_PWR_CLK (+3V3_VTX LDO EN through R84 1k), GPIO134 PA_NTC (ADC1, read in the
iStick window, erratum 3.11).
VTX LINES: series 1k R80/R81/R82 are drawn on the VTX sheet and placed at the ESP32 end
(back - feed limit with +3V3_VTX off, F21). GPIO13 VTX_AMP_PWM = ELRS vtx_omiso_pwm (RTC6705
PAOUT1, read PNP, not the SE5004-L bias), PA_DET is ADC2_CH0 (not readable with Wi-Fi on; not a
float).
+3V3 LDO: ELRS about 0.07 A in flight, Wi-Fi TX peaks 370 mA (DS Table 19, 802.11b 19.5 dBm) in
ELRS Wi-Fi mode only (disarmed, bench): +3V3 peak about 0.48 A with the FC and LED, inside the
LP5912 50-mA rating; LDO heat and the COUNT sum belong to the POWER sheet (P4-12).
PLACEMENT: SiP site from floorplan p3 (D55, D61); no 85 C halo rule (D69). Alternate part
ESP32-PICO-V3-02 (pins 28/29 unused here, P4-14).
```

# ELRS RGB LED

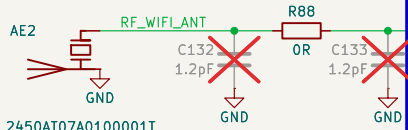
D4 XINGLIGHT XL-1010RGBC-2812B (copied from OpenRX-Lite-UFL) on +3V3, ELRS led\_rgb GPIO22, one LED (overlay ledidx\_rgb\_vtx [0]).

P4-15 DECIDED: KEEP ON +3V3. Datasheet: VDD 3.5–5.5 V, DIN VIH  $\geq$  2.8 V, 5 mA per colour with a 0.6 V sink knee. On +3V3 the supply is 0.2 V under the table minimum (blue / green may run below 5 mA; Vf + knee is about 3.4–3.6 V); DIN comes from a 3.3 V CMOS output on the same rail. On +5V the supply would be in range but the data guarantee is short (ESP32 VOH  $\geq$  0.8 VDD = 2.64 V at a high-impedance load, DS Table 18, vs VIH 2.8 V). Neither option sits fully inside the tables; the house RX circuit (OpenRX-Lite, -UFL, OpenAIO) runs this LED on 3.3 V, so it stays. Bench V-LED on the first boards: red, green, blue and white at 3.3 V, cold and hot. Fallback: VDD pin to +5V (one trace; C83 is 25 V).



D4 XINGLITE XL-1010RGBC-2812B (copied from OpenRX-Lite-UFL) on +3V3, ELRS led\_rgb GPIO22, one LED (overlay leddata\_rgb.vct [0]).

D5 15A: DECIDED: KEEP ON +3V3, Datasheet: VDD 3.5–5.5 V, DIM VIH >= 2.8 V, 5 mA per colour with a 100 nA knee. Knee is about 3.2 V under the table minimum. blue / green may run below 2.5 mA; V<sub>f</sub> + knee is about 3.4–3.6 V; DIN comes from a 3.3 V CMOS output on the same rail. On +5V the supply would be in range but the data guarantee is short (ESP32 V0H = 0.8 VDD = 2.64 V at a high-impedance load, DS Table 18, vs VIH 2.8 V). Neither option sits fully inside the tables; the house RX circuit (OpenRX-Lite, -UFL, OpenAO) runs this LED on 3.3 V, so it says. Bench V-LED on the first board: red, green, blue and white at 3.3 V, cold and hot. Failback: VDD pin to +5V (one trace; C83 is 25 V).

[illegible]

Wi-Fi (D58): LAN\_IN(2) -> RF\_WIFI 50 Ohm  
CPW <= 8 mm -> C133 shield (DNP) / R88 OF  
series (fitted) / C132 shield (DNP) -> AE2  
Johanson 2450AT07A01000011 (FEED 1/4, GND  
2/3). Same pin as Espressif Fig. 11 (C4/L1/C3  
and OpenRX-Lite-UFL (C22/L2/C21). The PIC0  
matches LAN\_IN to 50 Ohm inside, so the  
network only tunes the antenna: values from V7  
tuning (VNA at the feed or Johanson's service);  
R88 may become a Murata LQP03TN 0201  
inductor. The AE2 footprint carries the 5.0 x 3.2  
mm keep-out (DRU 'D21  
wifi\_antenna\_edge\_strip').

**ESP32-PICO-V3 MCU (ELRS)**

**Power Connections:**

- VDDA, VDD, VDDA3P3, VDDA3P3\_RTC, VDD3P3\_CPU, VDD3P3:** Connected to +3V3.
- C73 (1uF):** Connected to +3V3 and GND.
- C76 (100nF):** Connected to +3V3 and GND.
- R56 (10k):** Connected to +3V3 and GND.

**RF Connections:**

- RX\_EN (9):** Connected to RF\_WIFI.
- LNA\_IN (2):** Connected to RF\_WIFI.
- PA\_NTC (10):** Connected to RF\_WIFI.

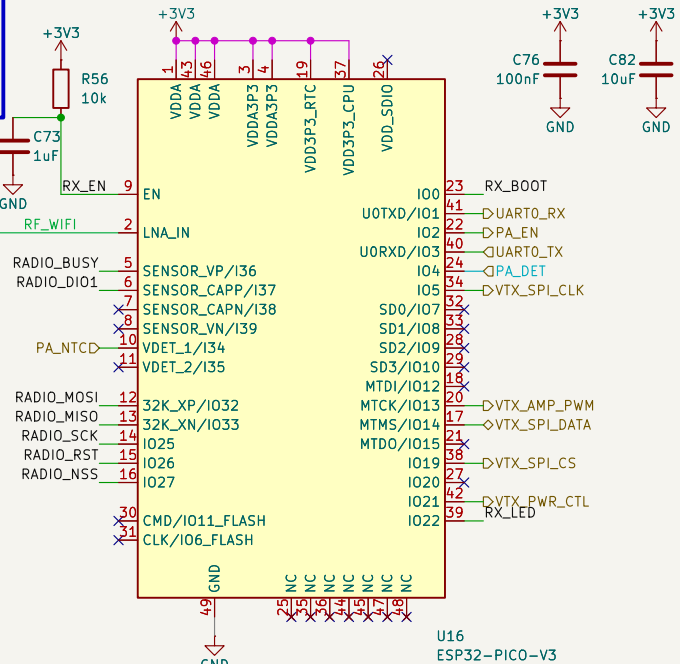
**UART Connections:**

- RX\_BOOT (23):** Connected to U16.
- UART\_RX (44):** Connected to U16.
- PA\_EN (22):** Connected to U16.
- UART\_TX (40):** Connected to U16.
- PA\_DET (24):** Connected to U16.
- VTX\_SPI\_CLK (34):** Connected to U16.
- VTX\_AMPL\_PWM (20):** Connected to U16.
- VTX\_SPI\_DATA (33):** Connected to U16.
- VTX\_SPI\_CS (38):** Connected to U16.
- VTX\_PWR\_CTL (42):** Connected to U16.
- RX\_LED (39):** Connected to U16.

**Other Connections:**

- SD0 (107), SD1 (108), SD2 (109), SD3 (1010), MTDI (1012), MTCK (1013), MTMS (1014), MTDO (1015), CMD (1011), CLK (1016):** Connected to GND.
- NC (15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40):** Connected to GND.

**U16 ESP32-PICO-V3**



# OpenDrone

Sheet: /RX/

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**Title: ELRS 2.4 GHz RX: ESP32-PICO-V3 + SX1281**

Size: A3

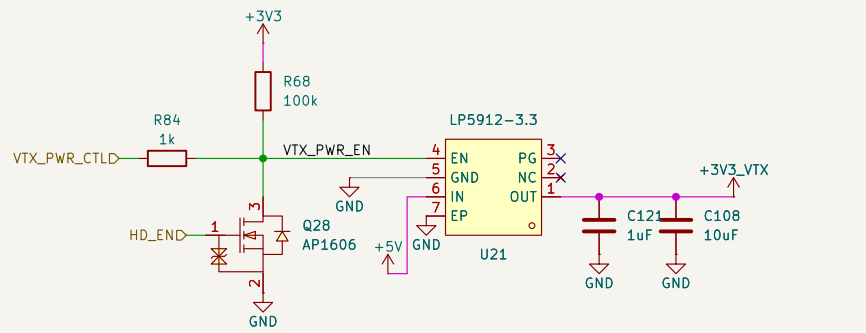
Date: 2026-10-09

Rev: rev1

KiCad E.D.A. 10.0.6

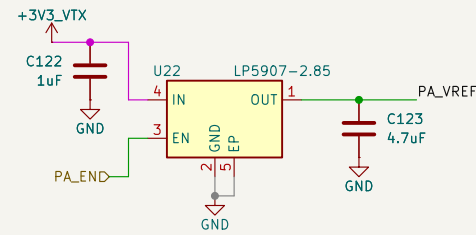
id: 11/14

## +3V3\_VTX LDO + HD GATE



U21 LP5912-3.3 on +5V; CIN = the shared 10 uF C13 on POWER between the U4 and U21 IN pins (within 2.4 mm of both). EN = VTX\_PWR\_EN; R68 100k to +3V3 (stock ELRS runs the VTX); R84 1k from ESP32 GPIO21 (patched-ELRS power-down hook); GPIO21 low gives VEN 0.03-0.04 V < VEN(OFF) 0.3 V; Q28 AP1606 drain = HD gate (HD\_EN high: <= 220 mOhm at VGS 2.5 V sinks 3.3 mA, VEN < 1 mV; HD\_EN before Betaflight drives PINIO1: <= 0.2 V vs VGS(th) 0.4 V min). COUT on +3V3\_VTX: C108 10u + C121 1u + C122 1u + 3k 100n + 2k 100p = 12.3 uF nominal, 3.7-7.0 uF effective at 3.3 V (!). Inside the LP5912 0.7-10 uF window. Place C108 at the RTC6705 VCO pins 39/40 (VCO bulk) and route +3V3\_VTX C108 -> U22 directly, else C122 becomes a 4.7 uF CL05A475MP5NRC. Load 0.10 A typical, about 0.17 A at the top PA step (root power table).

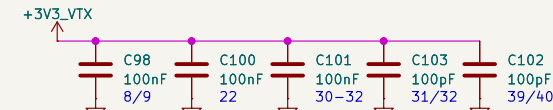
## PA REFERENCE LP5907-2.85



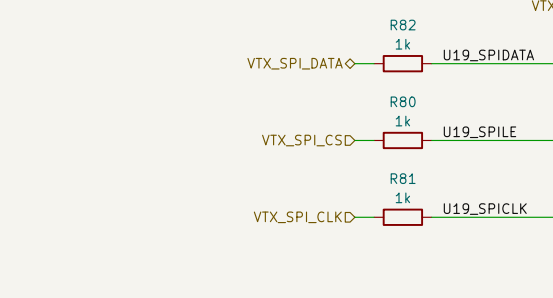
U22 LP5907-2.85 from +3V3\_VTX (TI SNV5798): HD mode removes PA\_VREF with the rail. EN = PA\_EN straight from ESP32 GPIO2 (strap: internal 45k pull-down + the LP5907 1 MOhm EN pull-down). C123 4.7 uF 0402 (> 0.7 uF effective); C122 1 uF (about 0.5 uF effective) only with the direct C108 -> U22 route. DEVIATIONS, bench-gated: (a) VREF window: 2.85 V +/- 2 % = 2.79-2.91 V vs SE5004L VREF H 2.80-2.90 V, and the +/- 2 % is specified only for VIN >= VOUT + 1 V (VIN here 3.3 V); V5 measures PA\_VREF on several units, cold and hot. (b) HD switch-on with ELRS holding PA\_EN high: +3V3\_VTX collapses before C123 discharges, so VOUT exceeds VIN + 0.3 V (LP5907 abs max, DS 5.1) by up to one body-diode drop (<= 0.7 V, < 1 ms, <= 10 uJ) from C123; V9 scopes PA\_VREF and +3V3\_VTX (< 0.5 V). Fallbacks: C123 down to 1 uF, or the patched ELRS drops PA\_EN on HD\_EN.

## RTC6705 VTX: 8 MHz CRYSTAL, SPI, VIDEO, PLL, PA DRIVE, PAOUT1 FEED

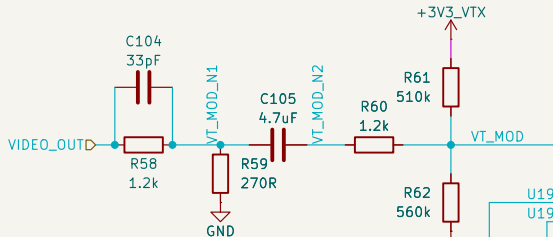
Decoupling at the RTC6705 supply pins (pin numbers next to each capacitor):



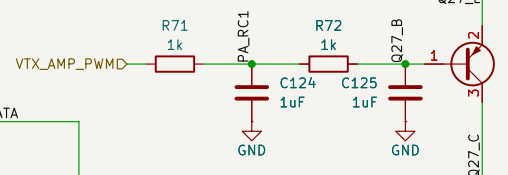
### VTX SPI 1k SERIES



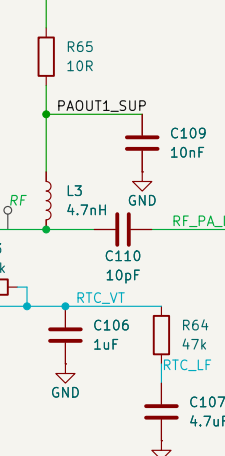
### VIDEO MODULATION NETWORK



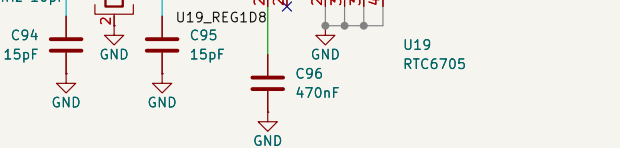
### PNP PA DRIVE



### PAOUT1 FEED



### PLL LOOP FILTER



RTC6705, checked pin by pin against OpenOSD-X v1.01: 23 REG108\_1 -> C96 470n (= ref); 26 REG108 open (ref DNI); 17 AVDD\_6.5 open (ref 3.3k + 1u, no 3.3 V; V5 checks for a 6.5 MHz spur); 31/32 BUFVDD/PAVDD -> +3V3\_VTX with C103 100p + C101 100n shared with 50; 40 LDD2V5 -> +3V3\_VTX direct with C102 100p (ref 51R + 1u filter; V5 video SNR); 39 VCOVDD with C102 + the 10 uF C108 placed there (ref 4.7u); 8/9 C98, 22 C100, Open: 1 SPLSE (internal pull-up = SPI mode), 2-4, 11-16 (11 RF\_VT2 audio input; V5 spur check), 18-21, 34, 36, 37, X4 exactly 8 MHz (ELRS R = 400). CL 10 pF = 15p + 15p in series + about 2.5 pF stray.

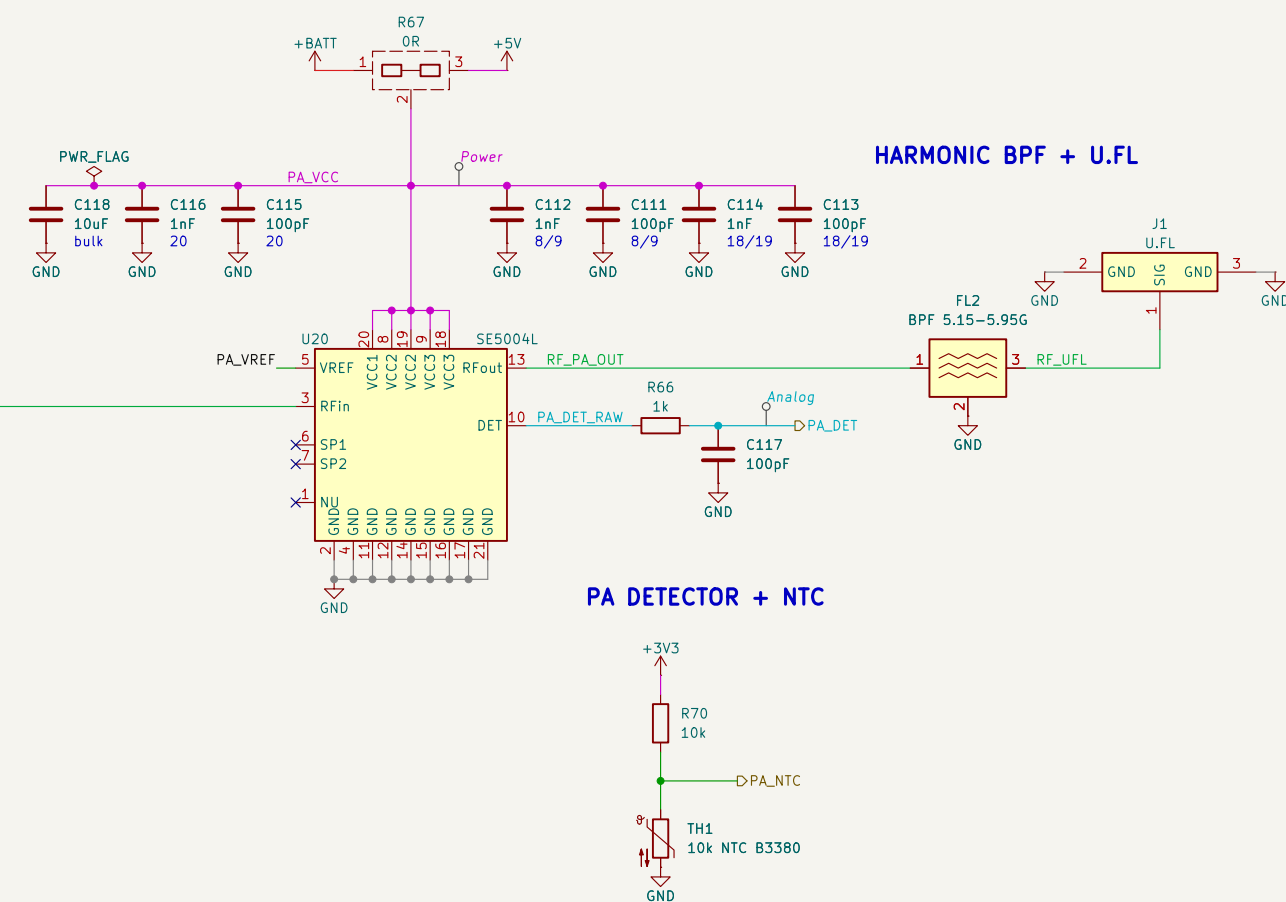
VTX SPI (PINMAP F21). R80 / R81 / R82 1k in series with CS / CLK / DATA, drawn here and PLACED AT THE ESP32 END of each line (bom\_plan): they limit back-feed into the unpowered RTC6705 in HD mode (<= 2.7 mA per high line; CS idles high -> +3V3\_VTX about 0.27 V on the LP5912 100 Ohm active discharge V9 limit 0.5 V; writes only at frequency changes). ELRS HSPI 1 MHz, mode 0, LSB first; tau = 1k + about 12 pF = 12 ns (26 ns 10-90 %) against >= 20 ns setup / hold in a 500 ns half period; the same RC on all three lines. SPIDATA is bidirectional (3-wire reads through R82, production test).

VIDEO + LOOP FILTER (OpenOSD-X v1.01). VIDEO\_OUT (OSD switch; 75R camera termination on the OSD sheet) -> R58 1.2k || C104 33p -> R59 270R shunt -> C105 4.7u -> R60 1.2k -> VT\_MOD; bias R61 510k / R62 560k from +3V3\_VTX (1.73 V) = OpenOSD-X R24 C17 R25 C21 R29 R30 R31; Load seen at VIDEO\_OUT: 1.47 kOhm (R58 + R59), about 0.27 kOhm above 5 MHz (C104 bypasses R58); the OSD level note uses It; Loop filter = OpenOSD-X; CF -> R63 100k -> VT -> C106 1u and R64 47k + C107 4.7u to GND. RTC\_CP / RTC\_VT / RTC\_LF and VT\_MOD\* are high-impedance nodes (Analog net class); short copper, >= 5 mm from the boost and the ESC phases.

PNP PA DRIVE, ESP32 GPIO13 VTXAMP\_PWM (ELRS vtx.amp\_pwm, 10 kHz) -> two RC poles (R71 / C124, R72 / C125, 1k / 1u each) -> BC857BM base; emitter R73 (Re) to +3V3\_VTX; collector Q27.C -> R65 10R -> PAOUT1\_SUP -> L3 -> RTC6705 PAOUT1; it sets the RTC6705 output-stage current, not the SE5004L bias. Inverting, current mode, ngspice (Nexperia BC857BM model, PAOUT1 sink model): Re 10R gives 32-59 mA at the top step over hFE 220-475 and -20.85 C, Q27 <= 114 mW (Ptot 130 mW at 85 C), base ripple 2.1 mV p-p, settles in 8 ms (ELRS 20 ms loop), GPIO13 floating at reset: 0 mA; HD mode: both junctions reverse-biased 3.3 V (VEBO 5 V). Steps per 6 dB at Re 10R: 11-13 at 1 mA, 16-19 at 2 mA, 25-32 at 4 mA; V8 measures the PAOUT1 current at 25 mW and V5 sets Re (27R gives >= 20 steps from 1.5 mA). Start value R73 = 10R.

PAOUT1 FEED + DC BLOCK (OpenOSD-X L4 C40 C43 R34, fed by the PNP stage instead of its NPN follower). PAOUT1 (pin 35, +2 dBm) -> L3 4.7 nH choke to PAOUT1\_SUP (C109 10 nF X7R) <- R65 10R <- Q27.C; C110 10 pF DC block into RF\_PA\_IN (SE5004L pin 3 'DC block required'); Only L3, C109 and C110 sit in the 5.8 GHz part-free band between pin 35 and the PA (50 Ohm CPWG < 5 mm, 45 degree). RF class on RF\_PAOUT1 / RF\_PA\_IN / RF\_PA\_OUT / RF\_UFL (pattern \*/RF-\*).

## SE5004L PA + SUPPLY, DETECTOR, BPF + U.FL



SE5004L PA (Skyworks 202393B + EK1). RfOut has an internal DC shunt to GND and only RFin is marked 'DC block required'; the BPF, U.FL and antenna carry no DC, so RfOut -> FL2 directly (internal output match, no match parts on the line; P5 EM-checks the line). PA\_VCC = R67 selector: one 0402 OR from COM to +BATT (default, pads 1-2) or +5V (pads 2-3, hand rework); the footprint anchor is the pad 1-2 midpoint, so the CPL centres the default part; VCC1/VCC2 <= VCC3 holds (one net). EK1 decoupling: 1n = 100p at 8/9, at 18/19 and at 20, on both package sides, + C118 10 uF shared, no ferrite. VREF from PA\_VREF (abs max 3.6 V). SP1/SP2 (optional EVM caps) and NU open. EP + pins 2, 4, 11, 12, 14-17 to GND with a via field.

PA DETECTOR + NTC (spec 4.8). DET (0.325 V no RF .. 1.0 V at 27 dBm) -> PA\_DET\_RAW -> R66 1k / C117 100p -> PA\_DET (ESP32 GPIO4, ADC2); the trace crosses sides, keep the RC at the ESP32 end of R66. TH1 NCP03XH103F05RL in the PA ring + R70 10k from +3V3 -> PA\_NTC (GPIO34, ADC1, input only); 1.65 V at 25 C, BPF + U.FL; FL2 Walsin RFBF1608060K9BQ1C on the PA side away from the PA body (power handling at 0.71 W; open item 07); J1 U.FL with L2 cut out under the signal pad (RF\_PAD\_CUTOUT). >= 4 GND vias at the tabs.

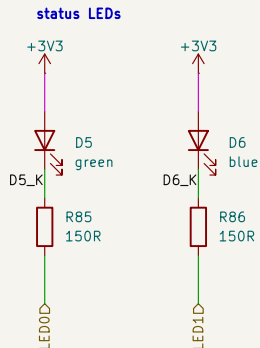
## SHEET NOTES

SHEET SOURCE: Generated by gen\_vtx.py (hardware/tools/sch\_build.py + sch\_root.py). No OpenDrone sibling sheet or commons block exists for an RTC6705 VTX, so the circuit follows the OpenOSD-X v1.01 reference schematic (RTC6705 supply pins, video network, loop filter, PAOUT1 feed), the Skyworks SE5004L datasheet 202393B + EK1 and DESIGN-SPEC 4.8. Refs, values, MPNs and footprints: hardware/bom\_plan.json. Signal flow: ESP32 controls (RX sheet) on the left, RTC6705 in the middle, PA, BPF and U.FL on the right.

BENCH ITEMS on this sheet: RTC6705 pin 11 RF\_VT2 open (V5 spur check); PA drive Re final value (V5 / V8, 27R fallback line); PA\_VREF window and the HD switch-on excursion (V5 / V9, PA REFERENCE note); VIDEO\_OUT OSD contrast and edges (V9, OSD sheet).

# OpenDrone

|                                          |                  |           |
|------------------------------------------|------------------|-----------|
| Sheet: /VTX/<br>File: vtx.kicad_sch      |                  |           |
| Title: 5.8 GHz VTX: RTC6705 + SE5004L PA |                  |           |
| Size: A2                                 | Date: 2026-10-09 | Rev: rev1 |
| KiCad E.D.A. 10.0.6                      | Id: 12/14        |           |



STATUS LEDs (D5 D6 R85 R86) source: OpenFC-Lite-Mini hardware/rp2350a.kicad\_sch 'status LED' block (R52 + R53 2x 75R, D2 XL-1005UBC), copied and adapted: sink drive per hardware/sch\_contract.json and research/PINMAP.md, 2x 75R -> one 150R 0201, second column for LED1.  
 Circuit: +3V3 -> LED -> 150R -> GPIO, the pin sinks (contract). Betaflight default polarity, low = on, no LEDx\_INVERTED: LED0\_PIN PA7 (GPIO7, green D5), LED1\_PIN PA26 (GPIO26, blue D6).  
 Datasheet check (XINGLIGHT XL-1005UGC / UBC, 0402): VF 2.7-3.3 / 2.6-3.4 V at 20 mA, IF max 20 mA, PD 70 mW, VR 5 V. At a few mA the InGaN VF is about 0.2-0.3 V lower (inferred), so with VOL <= 0.5 V at 4 mA drive (RP2350 D5) the LED current is about 0.5-5 mA, typ. 2-3 mA, hard ceiling (3.3 - 2.4) / 150R = 6 mA; LED, 0201 resistor (<= 5.4 mW of 50 mW) and GPIO (4 mA setting, 100 mA IOVSS sum) all inside their ratings; brightness at 1-3 mA is about 17-135 mcd green and 4-36 mcd blue (lv scaled from 350-900 / 80-240 mcd at 20 mA), enough for an indicator. Reverse bias is 0 V when the pin drives high (IOVDD = +3V3).  
 Reset state: RP2350 pads come up with the pull-down on (PDE = 1, 36-113 kOhm), so both LEDs may glow faintly (about 10-30 uA) until Betaflight drives the pins and in BOOTSEL mode. Accepted, cosmetic.  
 Placement (P5): bottom side, fc\_core region (bom\_plan); the R may sit at either end of the LED, slow signal (D33). Pins LED0/LED1 = root nets /LED0, /LED1 (RP2350A).  
 P4 closed here: bom\_plan R85/R86 'P4 checks brightness vs VF' (above). P4-14 fields: RC0201FR-07150RL is a stocked Yageo RC\_L part (Element14 4008725, RS 198-7568: 150R 1 % 50 mW 25 V); no LCSC number known, field left empty (contract open item A9-LCSC).

# OpenDrone

Sheet: /LED/  
 File: led.kicad\_sch

**Title: Status LEDs**

Size: A5 Date: 2026-10-09  
 KiCad E.D.A. 10.0.6

Rev: rev1  
 Id: 13/14



